

JOONA PIIRAINEN

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JOB EXPERIENCE

Adalyon, Helsinki, Finland

01/2024 – -/-

Software Engineering Software Developer

- Part of a team building a behavioural analytics platform from the ground up.
- Combined different machine learning techniques to accurately assess team dynamics.
- Worked across the entire technology stack, including data engineering, web development, and iOS development.
- Worked with technologies such as PyTorch, CoreML, Python, FastAPI, Microsoft Azure, Swift, C++ and TypeScript.

Reaktor, Helsinki, Finland

05/2021 – 01/2024

Software Engineering Software Developer

- Collaborated with numerous clients to resolve business-critical technical issues.
- Extensive experience helping critical software projects succeed in the Defence and Security domain.
- Part of both junior and senior recruitment teams. Conducted both technical and non-technical interviews.
- Worked with a variety of technologies, such as React, TypeScript, Node.js, Clojure, Java, ClojureScript, PostgreSQL and AWS.
- Got to take part in many trainings, such as Cyber Security, Certified Scrum Master and Consulting 101.

Puheet.com, Espoo, Finland

10/2020 – 05/2021

Software Engineering Software Developer

- Responsible for business critical platform components, such as the main backend system and multiple web front-end applications.
- Worked with a variety of technologies, such as React, Vue, TypeScript, PHP, Node.js, MySQL, Elasticsearch Linux and AWS.

SKILLS

- **Ways of Working:** Passionate about helping teams thrive and achieve project success. Extensive experience facilitating activities such as retrospectives and project kickoffs. Extensive knowledge of agile methodologies.
- **Server Engineering:** Experience developing reliable, scalable and high-throughput server-side systems with Clojure, Python, dotnet and Java. Love developing fast and correct solutions to hard algorithmic problems.
- **Frontend Development:** Expert in building beautiful and reliable client-side applications with React, Next.js, and Tailwind CSS.
- **Mobile Development:** Experience building iOS mobile applications with Swift, SwiftUI, and Core ML.
- **Program Languages:** **Multilingual and adaptable**, extensive experience in Kotlin, TypeScript, Clojure, Haskell, Python, Java, and C#. Comfortable working with Rust, C, C++, and Ocaml (in no particular order).
- **Cloud Technologies:** Extensive experience with cloud platforms such as Microsoft Azure and AWS. Skilled in developing and maintaining infrastructure pipelines using Infrastructure as Code (IaC) methodologies.
- **Programming Paradigms:** In-depth understanding of various programming paradigms, including functional, object-oriented, logic, and array programming. Recognize that different paradigms are suited to different problems, and there are no universal solutions.
- **Programming Practices:** Committed to delivering high-quality software with expertise in a range of testing and quality assurance techniques.
- **Programming Domains:** Experienced and passionate about programming domains such as algorithms, programming language development, web development, and computational methods in mathematics.

PROJECTS AND CONTRIBUTIONS

fp A small, weird and unpractical programming language. 

- fp is an unconventional programming language inspired by a paper by John Backus. John Backus described the language in his 1977 Turing Award lecture.
- For more information and numerous examples, please refer to the project's README.md.

v1 Shell scripting in TypeScript. 

- v1 is a tool for writing shell scripts in TypeScript, powered by deno.

aoc Solutions to Advent of Code problems implemented in various programming languages. 

- 2015, 2016, 2020, 2021, 2022, 2023.

MISCELLANEOUS

- As a hobby, I enjoy rock climbing, solving programming puzzles, and studying mathematics. Recently, I have been focusing on Algebraic Geometry and its computational aspects.